# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — MATHS UNIT TEST SET B

Chapters 1–4: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials, The World of Numbers, Exploring Algebraic Identities** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

## SECTION — A (Objective Type Questions) 12 × 1 = 12

*Choose the correct option and write it in your answer book.*

**1.** The reflection of the point $(-8, -1)$ in the $y$-axis is: 1

**(a)** $(8, -1)$ **(b)** $(-8, 1)$ **(c)** $(8, 1)$ **(d)** $(-1, -8)$

**2.** The distance between $A(0, 0)$ and $B(6, 8)$ is: 1

**(a)** 11 units **(b)** 9 units **(c)** 12 units **(d)** 10 units

**3.** Two vertices of a rectangle are $(1, 2)$ and $(5, 2)$ and one side has length 3 above the $x$-axis. The other two vertices are: 1

**(a)** $(1, -1)$ and $(5, -1)$ **(b)** $(1, 3)$ and $(5, 3)$
**(c)** $(1, 5)$ and $(5, 5)$ **(d)** $(4, 2)$ and $(8, 2)$

**4.** The point $(1, 8)$ ___ the line $y = 4x + 3$. 1

**(a)** is the origin **(b)** does not lie on the line
**(c)** lies on the line **(d)** is the $y$-intercept of the line

**5.** The value of $1x^2 + 2x - 3$ at $x = 2$ is: 1

**(a)** 7 **(b)** 2 **(c)** -3 **(d)** 5

**6.** The constant term of $9x^3 + 5x^2 - 8x - 10$ is: 1

**(a)** −8 **(b)** −10 **(c)** 9 **(d)** 5

**7.** A rational number in lowest terms has denominator $2 \times 5^2$. Its decimal expansion terminates after how many places? 1

**(a)** 3 **(b)** 1 **(c)** 4 **(d)** 2

**8.** The number $0.\overline{45}$ in the form $\frac{p}{q}$ is: 1

**(a)** $\frac{61}{110}$ **(b)** $\frac{5}{11}$ **(c)** $\frac{6}{11}$ **(d)** $\frac{11}{5}$

**9.** The average of $\frac{1}{2}$ and $\frac{3}{4}$ is a rational number lying between them. It is: 1

**(a)** $\frac{4}{5}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{5}{8}$ **(d)** $\frac{7}{8}$

**10.** Expand $(2x + 5)^2$: 1

**(a)** $4x^2 + 20x + 25$ **(b)** $4x^2 + 20x - 25$ **(c)** $4x^2 + 25$ **(d)** $4x^2 + 10x + 25$

**11.** $(x + 2)^3$ equals: 1

**(a)** $x^3 + 6x^2 + 12x + 8$ **(b)** $x^3 + 8$ **(c)** $x^3 + 2x^2 + 4x + 8$ **(d)** $x^3 + 6x^2 + 6x + 8$

**12.** The expression $9x^2 + 24x + 16$ can be factorised as: 1

**(a)** $(3x - 4)^2$ **(b)** $(3x + 4)^2$ **(c)** $(9x + 4)^2$ **(d)** $(3x + 8)^2$

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 20–30 words each.*

**13.** Find the coordinates of the midpoint of the segment joining $P(-5,\ 3)$ and $Q(1,\ 9)$. 2

**14.** Find the values of the linear polynomial $-2x + 6$ at $x = 1$ and $x = 4$. 2

**15.** Without long division, state whether $\frac{3}{8}$ has a terminating or a repeating decimal. Give a reason. 2

**16.** The temperature in Ladakh is 4 °C at noon and drops by 15 °C by midnight. Find the midnight temperature. 2

**17.** Expand $(3s - 4t)^2$ using a suitable identity. 2

## SECTION — C (Short Answer Type) — 6 × 3 = 18

*Answer in about 40–60 words each.*

**18.** The midpoints of the sides of a triangle are $D(5,\ 1)$, $E(6,\ 5)$ and $F(0,\ 3)$. Find the vertices. 3

**19.** A phone bought for ₹10,000 loses ₹800 every year. Find its value after 3 years, write $v$ in terms of $t$ and after how many years it becomes ₹4,000. 3

**20.** Show that $3 + \sqrt{2}$ is irrational, given that $\sqrt{2}$ is irrational. 3

**21.** The sum of three numbers is 10 and the sum of their squares is 38. Find $xy + yz + zx$. 3

**22.** Show that the triangle with vertices $A(-2,\ 0)$, $B(4,\ 0)$, $C(1,\ 4)$ is isosceles. 3

**OR**

Show that $A(1,\ 1)$, $B(5,\ 1)$, $C(5,\ 4)$, $D(1,\ 4)$ are the vertices of a rectangle. 3

**23.** A gym charges a fixed fee plus a rate per hour of court use: 10 h costs ₹800 and 15 h costs ₹1100. Find the relation $y = ax + b$ and the bill for 20 h. 3

**OR**

Find $a$ and $b$ in °C $= a$ °F$+b$, given that ice melts at 0 °C (32 °F) and water boils at 100 °C (212 °F). Then convert 98.6 °F to °C. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**24.** The vertices of a triangle are $A(0,\ 0)$, $B(8,\ 0)$, $C(4,\ 6)$. 5

(i) Find the lengths of the sides and state its type.

(ii) Find the midpoints $D, E, F$ of $BC, CA, AB$ and the perimeter of triangle $DEF$.

**OR**

(i) Find the distance between $P(-2,\ 3)$ and $Q(4,\ -5)$. 5

(ii) Hence find the midpoint of $PQ$ and verify that it is equidistant from $P$ and $Q$.

**25.** (i) Prove that $\sqrt{2}$ is irrational. 5

(ii) Hence show that $3\sqrt{2} - 5$ is irrational.

**OR**

**Case study:** The Ishango bone has a column of notches grouped as 11, 13, 17, 19. A merchant of Lothal gets 15 copper ingots for every 2 bags of spices. 5

(i) What do 11, 13, 17, 19 have in common? Name the next three numbers of this kind.

(ii) How many ingots will he get for 12 bags?

(iii) Are natural numbers closed under subtraction? Give an example.

## END OF QUESTION PAPER — Total: 50 Marks